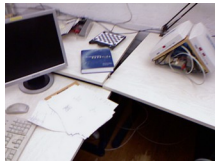
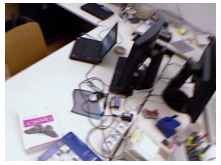
Pairs to a persistent 3D map



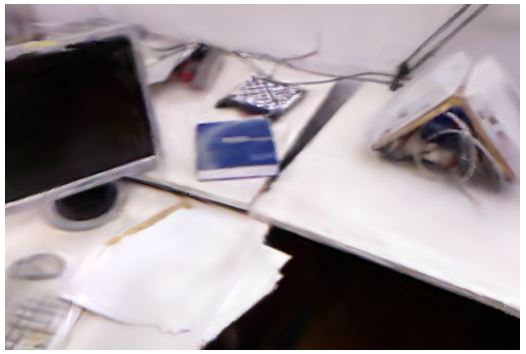


shared map–camera anchor



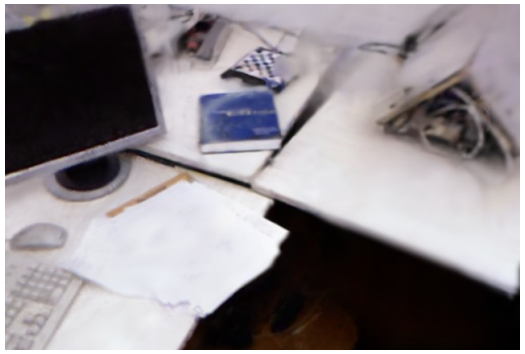
TUM desk: matched camera views

Photo-SLAM





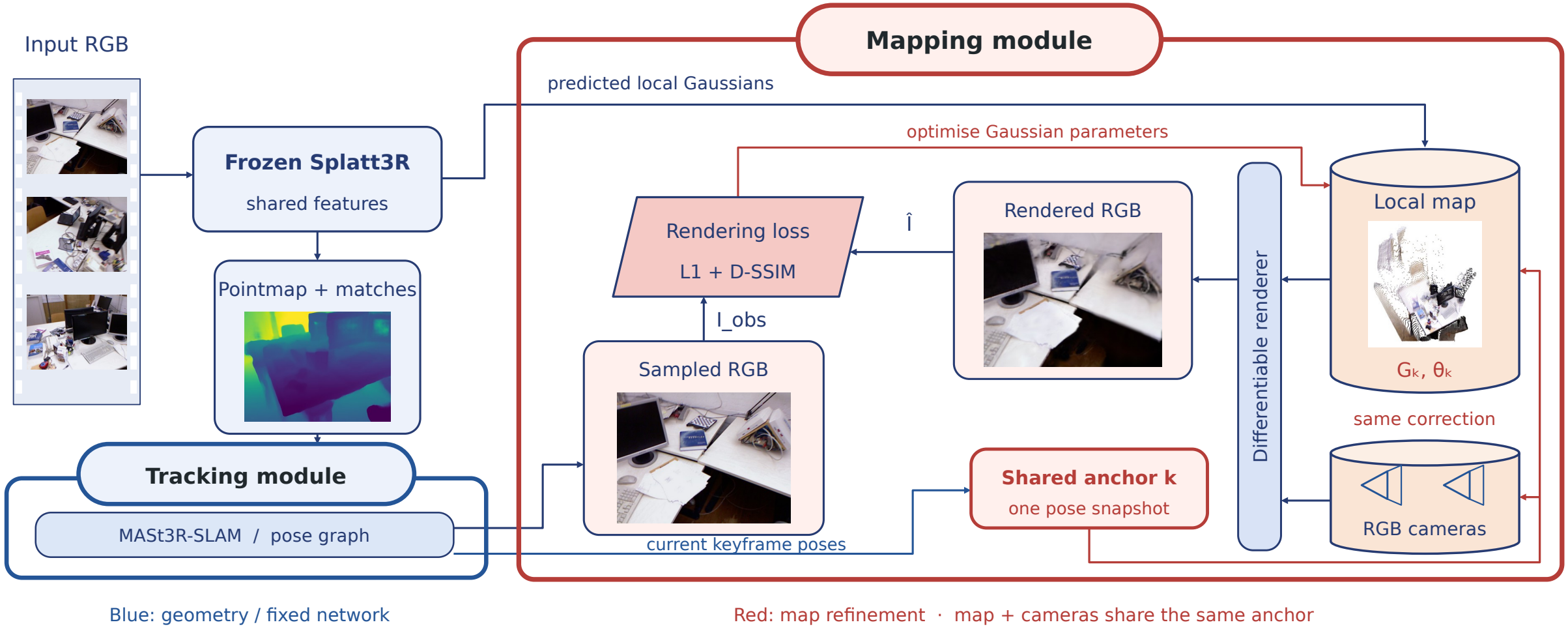
Ours





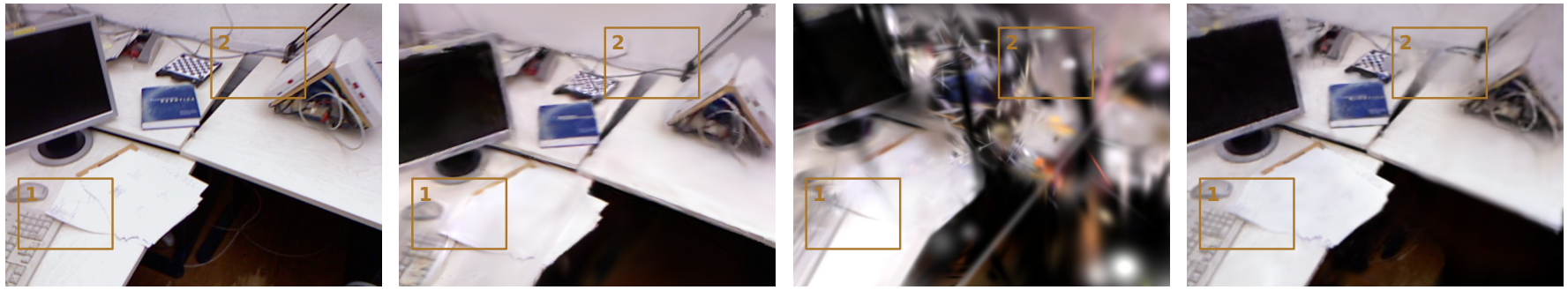


Mean paired-frame ΔPSNR: +3.53 dB · 100 frames


Input RGB
Frozen Splatt3R
shared features
Pointmap + matches
Tracking module
MASt3R-SLAM / pose graph
Mapping module
predicted local Gaussians
optimise Gaussian parameters
Rendering loss
L1 + D-SSIM
Î
Rendered RGB
I_obs
Sampled RGB
current keyframe poses
Shared anchor k
one pose snapshot
Differentiable renderer
Local map
Gk, θk
same correction
RGB cameras
Blue: geometry / fixed network
Red: map refinement · map + cameras share the same anchor

Real-data reconstruction and matched details

Ground truth | Photo-SLAM | MonoGS | Ours



Detail crops — method names are shown below each image



1 · GT | **2 · GT** | **1 · Photo-SLAM** | **2 · Ours** | **1 · MonoGS** | **2 · MonoGS** | **1 · Ours** | **2 · Photo-SLAM**

Additional real-data reconstructions

TUM fr1/360 | ETH3D table

Ground truth | Ours | Ground truth | Ours

